ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

MATHEMATICS

PRACTICE EXAM 3

MARKS: 100 **TIME: 2 hours**

Covers the full Grade 8 year with more problem solving: comparing cell phone deals, hire purchase with interest, exponents and integers, fractions in context, patterns, expressions and equations, a taxi-fare formula, geometry with parallel lines, triangles and quadrilaterals, Pythagoras, circles and cubes, a distance–time graph and probability.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of NINE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Show ALL your working. Answers alone will not get full marks.
4. A non-programmable calculator may be used, unless a question says otherwise.
5. Round answers to TWO decimal places where necessary. Diagrams are NOT drawn to scale.
6. Give reasons for your statements in geometry questions.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which of these numbers is NOT a prime number?

A 2 B 17 C 51 D 29

1.2 $(-1)^{100}$ = …

A 1 B −1 C −100 D 100

1.3 0,000 7 × 1 000 = …

A 0,07 B 7 C 70 D 0,7

1.4 $-3x(x - 2)$ = …

A $-3x^2 - 6x$ B $-3x^2 + 6x$
C $3x^2 - 6x$ D $-3x - 6$

1.5 If $y = 3x - 2$ and $x = -1$, then y = …

A 1 B 5 C −1 D −5

1.6 Each interior angle of an equilateral triangle is …

A 45° B 60° C 90° D 180°

1.7 A triangle with sides of 7 cm, 24 cm and 25 cm is …

A acute-angled. B obtuse-angled.
C right-angled. D equilateral.

1.8 The area of a circle with a radius of 10 cm is … (use π = 3,14)

A 314 cm² B 31,4 cm² C 62,8 cm² D 628 cm²

1.9 The simple interest on R500 at 10% per year for 2 years is …

A R50 B R100 C R600 D R1 000

1.10 The mean of 4 ; 8 ; 6 ; 10 ; 7 is …

A 6 B 8 C 35 D 7

[10]

QUESTION 2: FINANCE

2.1 Lerato compares two ways of getting a smartphone for two years:

Deal A (contract)	R350 per month for 24 months, phone included
Deal B (prepaid)	Buy the phone for R4 800 cash, plus R160 airtime per month for 24 months

2.1.1 Calculate the total cost of Deal A. (1)

2.1.2 Calculate the total cost of Deal B. (2)

2.1.3 Which deal is cheaper, and by how much? (2)

2.2 A laptop has a cash price of R9 000. On hire purchase, the buyer pays a 15% deposit. The balance plus simple interest at 12% per year is paid back in equal monthly instalments over 2 years.

2.2.1 Calculate the deposit. (1)

2.2.2 Calculate the interest on the balance. (2)

2.2.3 Calculate the monthly instalment. (2)

2.2.4 Calculate the total hire-purchase price of the laptop. (2)

[12]

QUESTION 3: EXPONENTS AND INTEGERS

3.1 Simplify: (a) $(3x^2)^3 \times 2x$ (b) $\frac{12a^5b^3}{4a^2b}$ (c) $(-2)^3 \times (-1)^5 + 5^0$ (6)

3.2 Solve for x: (a) $2^x = 32$ (b) $4^x = 64$ (2)

3.3 At 12:00 the temperature in Bethlehem is 7 °C. It drops by 2 °C every hour for 6 hours. What is the temperature at 18:00? (2)

[10]

QUESTION 4: FRACTIONS AND DECIMALS

4.1 A recipe needs $\frac{3}{4}$ cup of sugar for 6 people. How much sugar is needed for 10 people? Give your answer as a mixed number. (3)

4.2 Calculate WITHOUT a calculator: (a) $1\frac{2}{5} \times 2\frac{1}{2}$ (b) $(0{,}3)^2 + \sqrt{0{,}49}$ (c) Write $\frac{7}{8}$ as a percentage. (5)

[8]

QUESTION 5: PATTERNS, EXPRESSIONS AND EQUATIONS

5.1 Consider the pattern 1 ; 4 ; 9 ; 16 ; … (a) Write down the general term. (b) Calculate T_{12}. (c) Which term is equal to 400? (3)

5.2 Simplify: (a) $2a(3a - 4b) - a(a - 8b)$ (b) $(15x^3 - 10x^2 + 5x) \div 5x$ (5)

5.3 Solve for x: (a) $3(2x - 1) = 4(x + 2)$ (b) $\frac{x}{2} + \frac{x}{3} = 10$ (6)

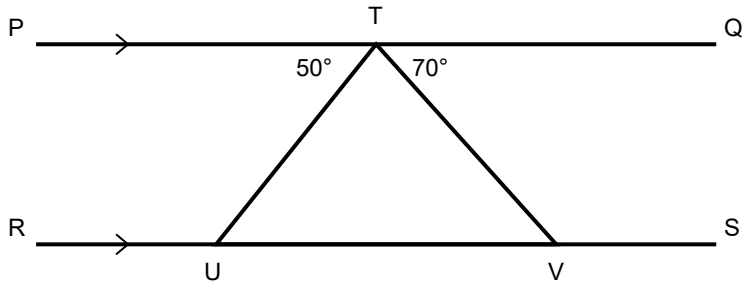
5.4 The sum of three consecutive whole numbers is 87. Write an equation and solve it to find the three numbers. (3)

5.5 A taxi company charges a fixed fee of R15 plus R8 per kilometre. The cost (C) for a trip of d km is $C = 8d + 15$. (a) Calculate the cost of a 12 km trip. (b) How far can you travel for R95? (3)

[20]

QUESTION 6: GEOMETRY

6.1 In the diagram, PQ ∥ RS. T is a point on PQ, and U and V are points on RS. ∠PTU = 50° and ∠QTV = 70°.



Calculate, giving reasons: (a) ∠TUV (b) ∠TVU (c) ∠UTV (6)

6.2 In ΔABC and ΔDEF, AB = DE, ∠B = ∠E and BC = EF. Are the triangles congruent? Give the condition. (2)

6.3 ABCD is a parallelogram with $\angle A = 3y + 15°$ and $\angle C = 5y - 25°$. Calculate y, ∠A and ∠B. Give reasons. (4)

6.4 In ΔXYZ, the exterior angle at Z is 120° and ∠X = 2∠Y. Calculate the sizes of ∠X and ∠Y. (3)

6.5 Calculate the size of each interior angle of a regular hexagon. (1)

[16]

QUESTION 7: PYTHAGORAS AND MEASUREMENT

7.1 A rectangular sports field is 40 m long and 30 m wide. Kabelo walks diagonally across the field instead of walking along two sides. How many metres does he save? (3)

7.2 An isosceles triangle has two equal sides of 10 cm and a base of 12 cm. Calculate (a) its perpendicular height and (b) its area. (3)

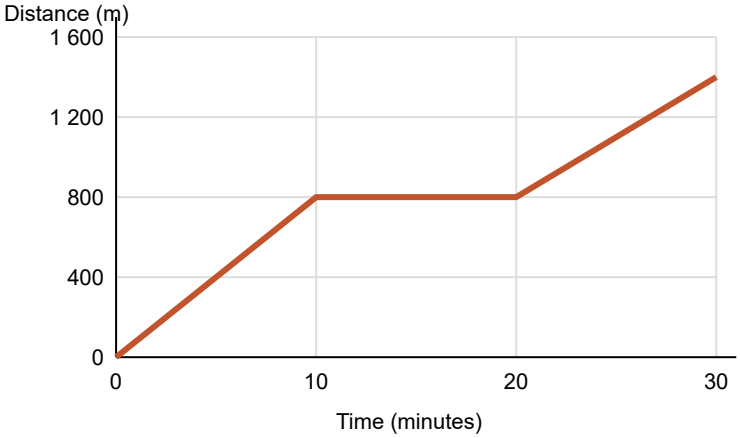
7.3 A round flower bed has a radius of 3,5 m. Use $\pi = \frac{22}{7}$ to calculate (a) its area and (b) the length of edging needed to go around it. (4)

7.4 The total surface area of a cube is 150 cm². Calculate the volume of the cube. (4)

[14]

QUESTION 8: GRAPHS

8.1 The graph shows Sipho's walk from home to his friend's house.



8.1.1 How far had Sipho walked after 10 minutes? (1)

8.1.2 What do you think happened between 10 and 20 minutes? (1)

8.1.3 How far is his friend's house from his home? (1)

8.1.4 Calculate his average speed in m/min during the first 10 minutes. (2)

8.1.5 During which time period did he walk faster: the first 10 minutes or the last 10 minutes? (1)

[6]

QUESTION 9: PROBABILITY

9.1 A spinner has eight equal sections numbered 1 to 8. It is spun once. Determine the probability of getting: (a) an even number; (b) a prime number; (c) a multiple of 3. Simplify your answers. (4)

[4]

TOTAL: 100

END OF QUESTION PAPER